

Introduction

The healthcare system is currently under significant pressure and this project introduces intelligent **virtual agents**, trained on medical resources, to provide immediate support where it is most needed. Rather than replacing human judgment, these agents **simplify and streamline critical workflows**, helping staff deliver quality care while preserving the human aspect of healthcare.

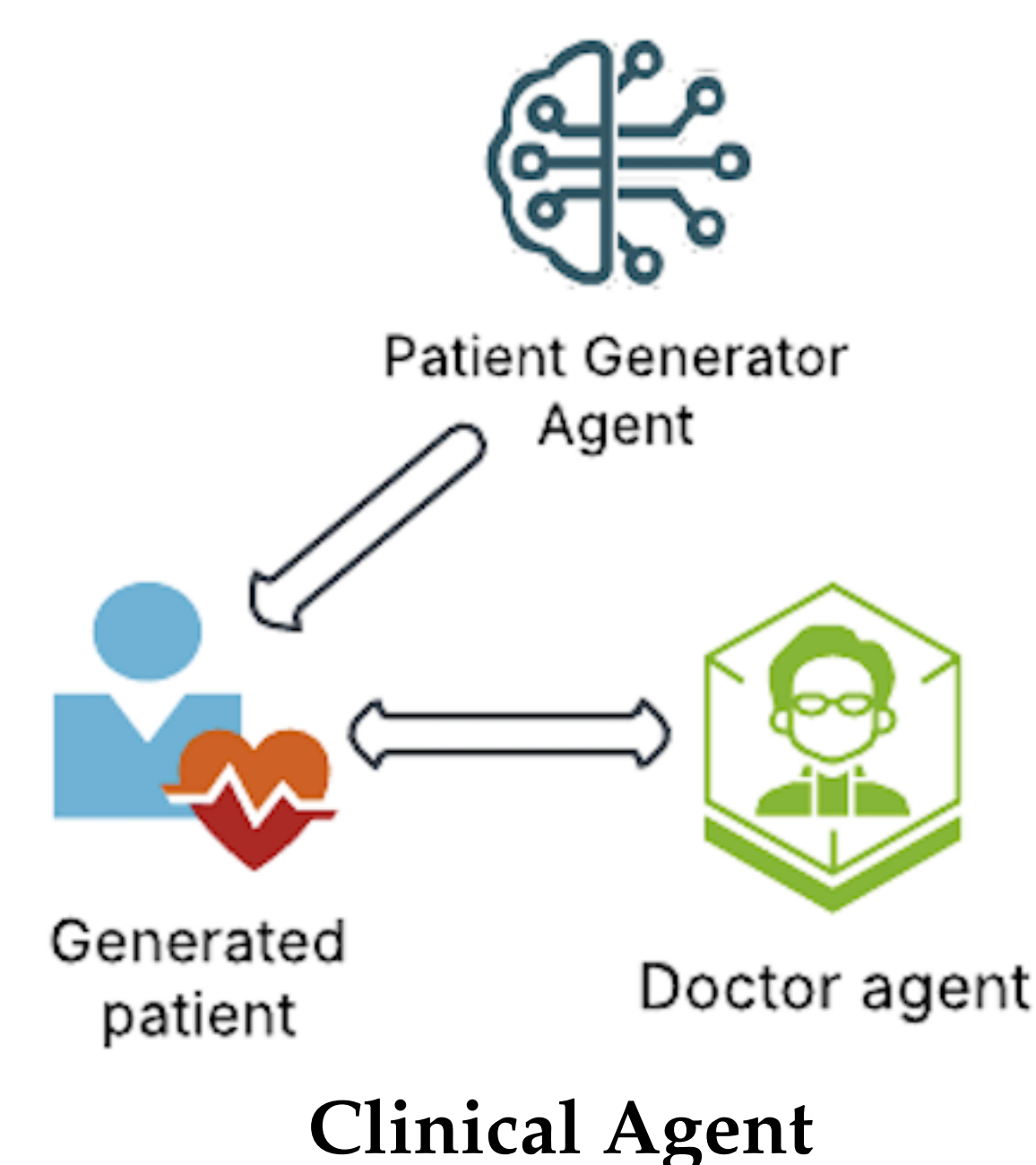
1. **Reduce Info-Santé Overload** — Slash routine call volume
2. **Support Medical Staff** — Rapid clinical recall, fewer omissions
3. **Assist Emergency Triage** — Assist or automate patient sorting

The goal of our project is to analyze the effects and performances of doctor agents utilizing techniques from state of the art *Agent Hospital* but using lower scale AI. Insights from our analysis contribute to the improvement of more modest medical agent models, which is critical for the development of cutting-edge models afterwards.

Agent Hospital

This work builds on the *Agent Hospital: A Simulacrum of Hospital with Evolvable Medical Agents* project, a complete hospital simulation where patients, nurses and doctors powered by LLMs interact and evolve through extensive autonomous simulations. It reports that after enough time in the simulation, **the doctor agents outperform state-of-the-art medical agents** on the *MedQA* benchmark.

Simplification - Our Clinical Agent



- 2 agents (Doctor Agent + Patient Generator Agent)
- Asynchronous simulation (more like data augmentation / self-supervised learning)
- Multiple choices task instead of language modeling
- Fine-tuning instead of information retrieval for simulation learning
- Usage of lower scale models: *PubMedBERT* and *FLAN-T5-base*

Challenges

Developing this project from scratch came with several challenges:

- Time and implementation access constraints
- Limited dataset
- Model capacity limits for training and generation

Data and Annotation

```
"Full name": "Amina Patel",
"Age": "45",
"Gender": "Female",
"Symptoms": "Warmth, tenderness, pain, and swelling in her right calf; occasional skin redness",
"Symptom duration": "Starting a week ago, the symptoms have become more pronounced over the past few days",
"Recent travel history": "Amina recently took a 12-hour flight from Mumbai to New York for a business conference, followed by a 3-day drive to her client's location in Boston",
"Medical history": "Amina has been diagnosed with a blood disorder that increases her risk of developing clots; she is currently on medication to manage the condition but has occasionally missed doses due to work commitments",
"Mixed diseases": [
  "Thoracic Outlet Syndrome",
  "Artificial Limbs",
  "Marfan Syndrome",
  "Deep Vein Thrombosis",
  "High Blood Pressure"
],
"Correct disease": "Deep Vein Thrombosis",
"Diagnosis cause": "Vein thrombosis",
"patient_number": "0003677"
```

Generated Agent Patient

- Data sourced from MedlinePlus
- Processed with Llama-2 and Mistral-7B to generate 5530 synthetic cardiology patient profiles



Methodology

- Simulation and evaluation strategies: full fine-tuning
- Pre-training on textbooks strategy: masked language modeling
- Baseline base models: PubMedBERT and FLAN-T5-base
- Doctor Agent base model: PubMedBERT

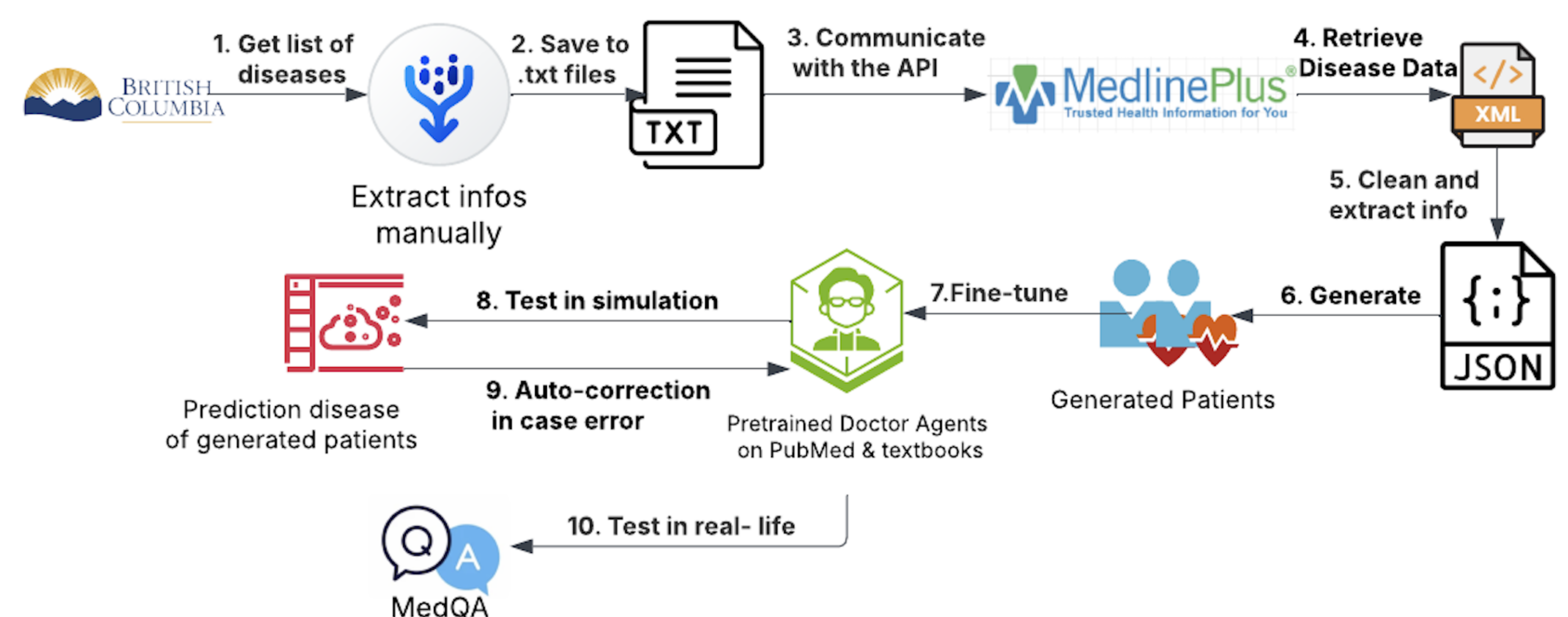
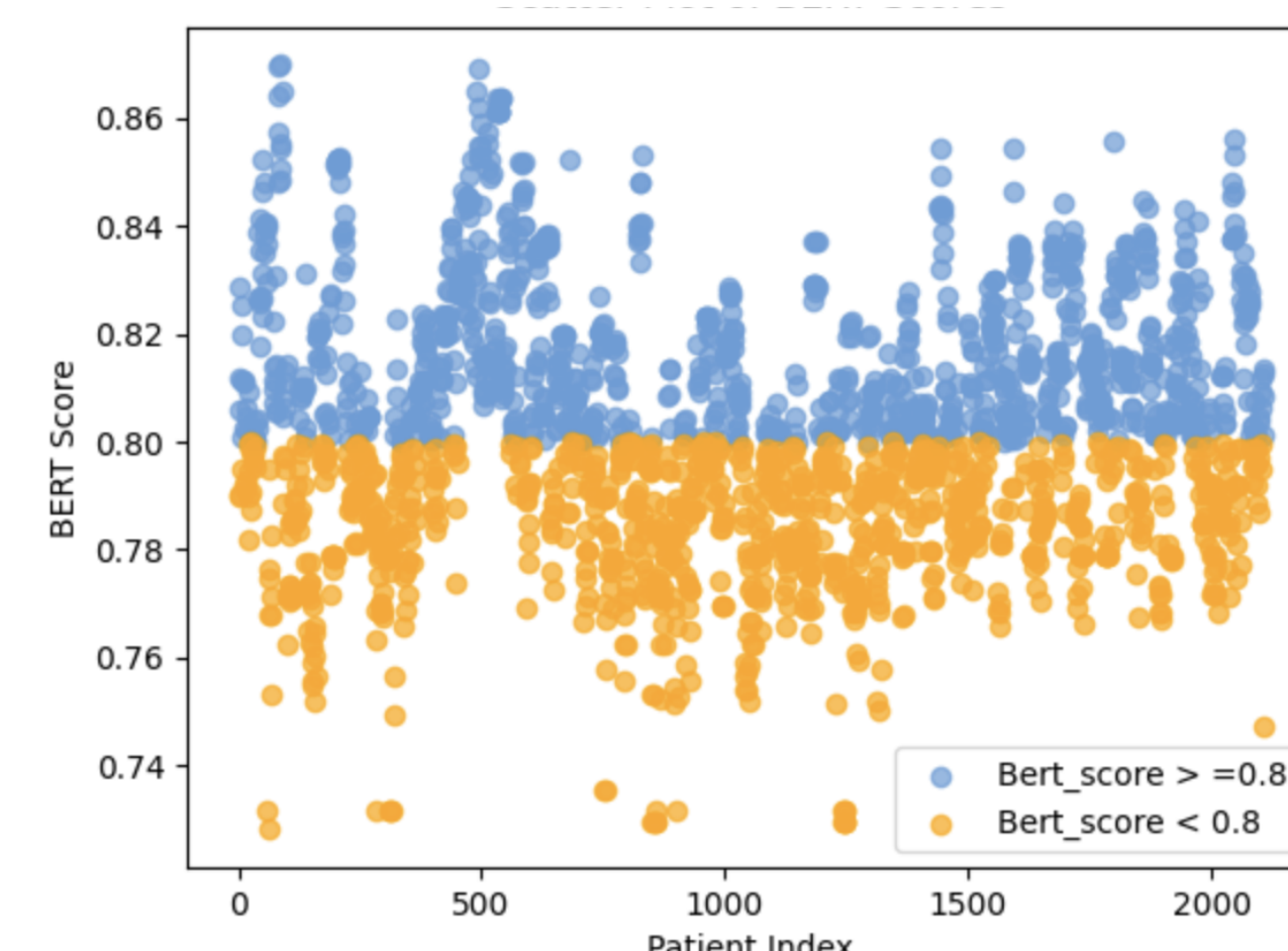


Figure 1: AI Doctor Agent Training Workflow.

Results



- BERTScore evaluation using the *yikuan8/Clinical-Longformer* model
- Evaluation conducted on 2110 patients, comparing only the symptom field and the full summary field

BERTScores for Synthetic Patients

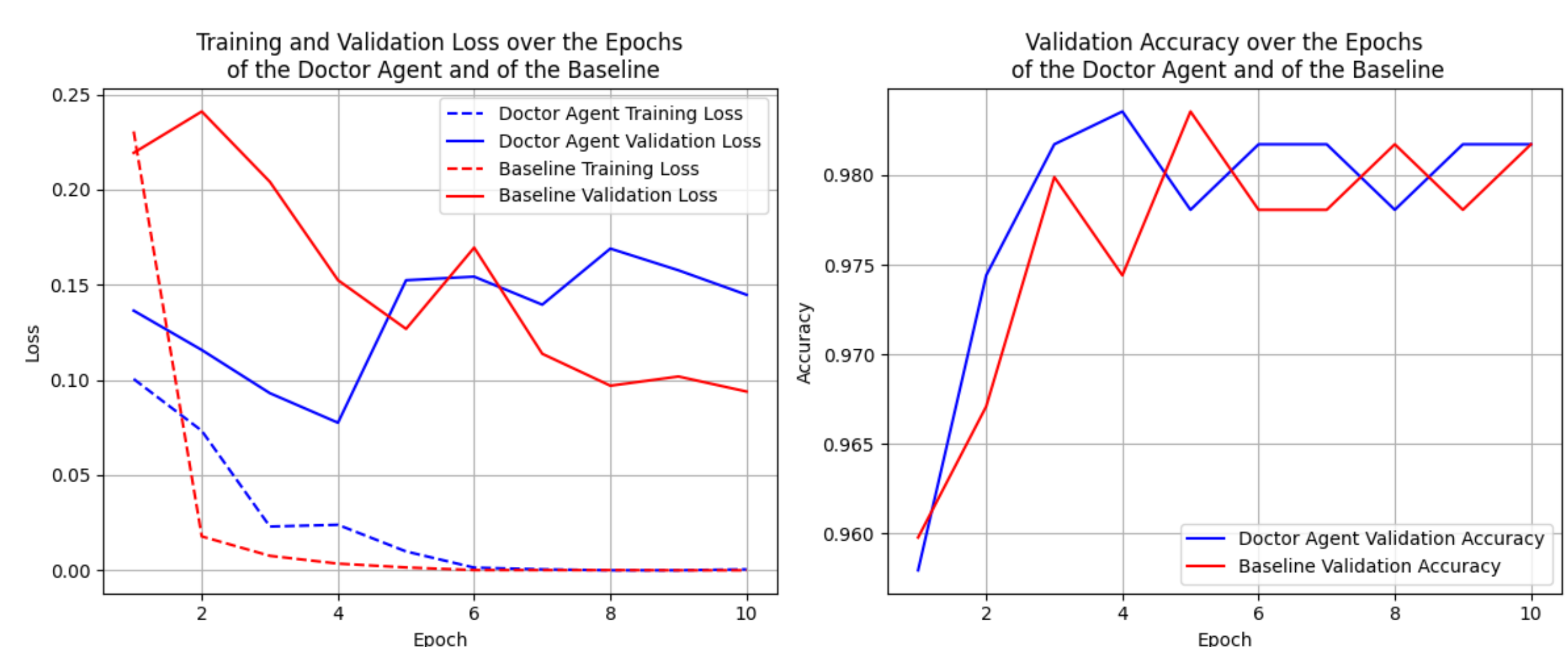


Figure 2: Virtual evaluation of the doctor agent and baseline for the **disease identification task** (on generated patients). The baseline used is PubMedBERT with extra pre-training on textbooks and fine-tuning on MedQA.

	Best valid accuracy	Test accuracy
PubMedBERT baseline	0.3357	0.3299
FLAN-T5-base baseline	0.2461	N/A
PubMedBERT + textbooks baseline	0.3239	0.3543
Doctor agent	0.3160	0.3142

Table 1: Real-life evaluation of baselines and of the doctor agent on the **MedQA test**: advanced medical multiple-choice questions.

Analysis and Conclusion

- Medical pre-training is necessary to perform on MedQA (FLAN-T5-base)
- The simulation fails to improve the doctor agent on MedQA, as it lacks sufficient difficulty due to mismatched disease distributions and limited diversity restricted to a single department (e.g., cardiology)
- The generated patients may introduce excessive noise or fail to contribute meaningful knowledge to enhance the doctor agent's prior understanding
- Full fine-tuning is not a good learning method for the simulation and for the baseline on MedQA (catastrophic forgetting)

Future work

Short-term

- Enhance predictive accuracy and benchmark against LLAMA 1B/7B
- Adjust fine-tuning and pre-training methods (e.g., LoRA, prefix-tuning, partial fine-tuning) and perform broader hyperparameter search
- Complexify the simulation task by diversifying patient profiles and evaluating them using another LLM

Long-term

- Building a *robust evolving dataset*
- Using advanced models like GPT-4 or + for training doctor agent
- Adding a *chat-based interface*
- Enabling *phone and voice interaction*